

Cisco Codec EQ Manual

1. Official Cisco Sources

This manual is based on official Cisco documentation for the Cisco Room Kit EQ and RoomOS endpoints, especially the Cisco Room Kit EQ Data Sheet, the Cisco Room Kit EQ Installation Guide, the Cisco Room Kit EQ support and documentation page, and the Cisco Board, Desk, and Room Series Administration Guide for RoomOS.

The source manual identifies the Cisco Codec EQ as the central collaboration codec in the Cisco Room Kit EQ bundle. It runs RoomOS and connects displays, cameras, microphones, touch controllers, USB peripherals, and network services.

Practical setup, verification, troubleshooting, and maintenance notes were added to make this manual more useful for installation and support work. Exact options may vary depending on RoomOS version, registration method, connected peripherals, licensing, Webex/CUCM/Expressway deployment, and room design.

2. Device Description

The Cisco Codec EQ is the video conferencing codec and collaboration compute engine used in the Cisco Room Kit EQ system.

It acts as the central processing unit for a meeting room or large collaboration space. It manages video, audio, content sharing, displays, cameras, microphones, touch controller interaction, calling, network registration, and room collaboration workflows.

The Codec EQ is commonly used in:

- Meeting rooms
- Boardrooms
- Training rooms
- Classrooms
- Auditoriums
- Hybrid collaboration rooms
- Executive conference spaces
- Large video conferencing spaces

In a smart office or meeting room topology, the Codec EQ should be treated as the **central video collaboration hub**. It connects the room display system, camera system, microphone system, network, and user control interface into one integrated conferencing endpoint.

The device can register to:

- Webex
- Cisco Unified Communications Manager
- Cisco Expressway-based deployments

The Codec EQ is not a general-purpose PC, network switch, or router. Its role is video collaboration, media processing, room device integration, and conferencing control.

3. Technical Specifications

The Cisco Codec EQ is manufactured by Cisco and is part of the Cisco Room Kit EQ collaboration system.

Main characteristics include:

- RoomOS operating system
- Video conferencing codec and collaboration engine
- HDMI inputs and outputs
- Support for up to three external displays
- HDMI camera input
- USB-C laptop content and passthrough support
- USB peripheral support
- Analog audio connections
- Ethernet LAN
- Wi-Fi support on non-NR models
- PoE++ support on the first four Ethernet peripheral ports
- Room Navigator touch controller support
- HTTPS web interface
- SSH access
- xAPI control
- Cisco Control Hub management
- Webex registration support
- CUCM registration support
- Expressway registration support
- Wireless sharing through Webex, Miracast, and AirPlay
- SRTP media encryption
- Kensington security lock

Power:

Input power: 100–240 VAC, 50/60 Hz
Average power draw: About 20 W

PoE information:

First four Ethernet ports support PoE++ / 802.3bt
Maximum: 90 W per port
Combined budget across first four ports: 90 W

Physical information:

Width: 18.6 in / 47.3 cm
Height: 1.7 in / 4.4 cm
Depth: 8 in / 20.2 cm
Weight: 5.1 lb / 2.33 kg

Operating environment:

Temperature: 0°C to 35°C / 32°F to 95°F
Relative humidity: 10% to 90%

Mounting options:

- Wall mounting
- Rack-ear mounting
- Free-standing placement

Important note:

NR models do not support Wi-Fi. For network planning, confirm whether the installed model is a Wi-Fi-capable model or an NR model.

4. Ports, Buttons, and Indicators

The Cisco Codec EQ includes several connection types used for displays, cameras, network, peripherals, audio, USB devices, and power.

4.1 Ethernet LAN and Peripheral Ports

The codec includes RJ-45 Ethernet ports for network and IP peripheral connectivity.

They are used for:

- LAN connection
- Webex/CUCM/Expressway registration
- IP peripherals
- Touch controller connection
- Camera or room peripheral connectivity
- PoE-powered peripheral devices

The first four Ethernet ports support PoE+ + power delivery with a combined power budget. This is important when powering connected cameras or peripherals.

4.2 HDMI Inputs

HDMI inputs are used for video sources such as cameras or content sources.

Important camera note:

- The HDMI camera should be connected to HDMI input 1.

If a camera is connected to the wrong input, the codec may not detect it as expected or may not provide the expected video source behavior.

4.3 HDMI Outputs

HDMI outputs are used to connect external displays.

The Codec EQ can support up to three displays depending on room design and configuration.

Displays may be used for:

- Far-end video
- Local presentation
- Content sharing
- Meeting interface
- Dual-screen or triple-screen layouts

4.4 USB-C Port

USB-C supports laptop content and passthrough workflows.

It may be used for:

- Laptop content sharing
- USB-C passthrough
- 4K content sharing where supported
- Bring-your-own-device workflows depending on deployment

4.5 USB Ports

Additional USB ports support compatible peripherals.

USB devices may include:

- Cameras
- Audio peripherals
- Touch or control accessories
- USB-connected room devices

Compatibility should always be checked against Cisco documentation and RoomOS support.

4.6 Analog Audio Connectors

Analog audio connectors support microphone and speaker integration where required.

They may be used for:

- External microphones
- Audio input
- Speaker output

- Integration with room audio systems

4.7 AC Power Inlet and Power Switch

The codec uses an AC power inlet and includes a power switch.

If the system does not power up, verify both:

- AC power cable
- Codec power switch

Some connected peripherals may also have their own power requirements.

4.8 Kensington Security Slot

The Kensington security slot provides a physical security option for preventing unauthorized removal.

4.9 Indicators and On-Screen Status

The codec reports operational state through:

- On-screen messages
- Room Navigator interface
- Web interface
- Control Hub status
- LEDs or connected display status where available

During troubleshooting, always check the on-screen status and management interface before replacing hardware.

5. Installation and Setup Notes

Before installation, confirm the codec model, room design, display count, camera model, microphone plan, network registration method, cable lengths, mounting method, and power availability.

Basic installation steps:

1. Place or mount the codec in the planned location.
2. Maintain at least 10 cm / 4 in. clearance at the front and rear.
3. Do not block ventilation openings.
4. Connect AC power using a compatible power cable.
5. Confirm the power switch is in the correct position.
6. Connect the primary display to an HDMI output.
7. Connect additional displays if used.
8. Connect the HDMI camera to HDMI input 1.
9. Connect microphones or audio peripherals.
10. Connect the Room Navigator or controller.
11. Connect Ethernet LAN or configure Wi-Fi if supported.
12. Connect USB-C or USB peripherals if used.
13. Power on the system.

14. Follow the on-screen setup process.
15. Upgrade RoomOS software before production use if recommended.
16. Register the device to Webex, CUCM, or Expressway.
17. Test calls, audio, video, content sharing, and room controls.
18. Document all connections and management information.

Important installation notes:

- Do not use a VISCA cable for camera control.
- The PTZ 4K camera can be powered through PoE on supported ports or with the specified power supply.
- Do not exceed the PoE budget across the first four ports.
- Use the correct display input on the room screen.
- Keep the system ventilated to prevent overheating.
- Select the correct network model, especially for NR vs non-NR Wi-Fi support.

6. Network and Registration Setup

The Codec EQ must connect to the network for calling, registration, remote management, updates, and cloud services.

Supported registration options include:

- Webex
- Cisco Unified Communications Manager
- Cisco Expressway

6.1 Ethernet Network

Ethernet LAN is the preferred network connection for stable conferencing.

Check:

```
Ethernet cable connected
Switch port active
Correct VLAN
DHCP or static IP configured
Gateway reachable
DNS working
Firewall rules allow required traffic
```

6.2 Wi-Fi Network

Wi-Fi is supported on non-NR models.

Wi-Fi may support:

- WPA3
- WPA3-Enterprise

- Wireless content sharing workflows depending on configuration

Important note:

NR models do not support Wi-Fi. If Wi-Fi options are missing, confirm the exact hardware model.

6.3 Registration Checks

After network setup, verify:

```
Device receives IP address
Device reaches required services
Registration status is online
Date and time are correct
Certificates are valid
Room Navigator shows normal status
Control Hub or call platform shows endpoint online
```

If registration fails, check network connectivity, DNS, firewall, time settings, certificates, and registration credentials.

7. Display and Camera Setup

The Codec EQ depends on correct HDMI and camera wiring.

7.1 Display Setup

Connect displays to HDMI outputs.

If the display is blank:

- Confirm display power.
- Confirm correct HDMI input is selected.
- Confirm HDMI cable is seated.
- Test another HDMI cable if needed.
- Check codec output settings.
- Confirm the display supports the selected resolution.

7.2 Camera Setup

Connect the HDMI camera to input 1.

If using a PTZ camera:

- Power it through supported PoE ports or specified power supply.
- Do not use a VISCA cable for camera control.
- Confirm network/IP peripheral connection if applicable.
- Confirm camera is detected by the codec.
- Check orientation if mounted inverted.

If the camera image is upside down, configure the codec to flip the camera image according to the installation and administration guidance.

8. Content Sharing and USB-C Operation

The Codec EQ supports wired and wireless content sharing.

Content sharing options may include:

- USB-C laptop content
- Webex wireless sharing
- Miracast
- AirPlay

8.1 USB-C Sharing

Use USB-C for laptop content and passthrough where supported.

Check:

```
USB-C cable supports required function
Laptop supports video output
Codec detects USB-C source
Correct sharing mode is selected
Display layout is correct
```

8.2 Wireless Sharing

Wireless sharing may use Webex, Miracast, or AirPlay depending on deployment.

If wireless sharing fails, check:

- Feature enabled in configuration
 - Device on correct network
 - Firewall and network discovery
 - RoomOS version
 - Client device compatibility
 - Wi-Fi availability on model
 - Enterprise security restrictions
-

9. Room Navigator and Management Access

The Codec EQ is controlled and managed through several interfaces.

Management methods include:

- Cisco Room Navigator
- HTTPS web interface

- SSH
- xAPI
- Cisco Control Hub

9.1 Room Navigator

The Room Navigator is the primary in-room control interface.

It can be used for:

- Starting and ending calls
- Joining meetings
- Sharing content
- Controlling camera
- Adjusting volume
- Accessing room controls
- Viewing room status

9.2 Web Interface

The HTTPS web interface allows administrators to configure and manage the endpoint.

It may be used for:

- Device settings
- Diagnostics
- Network configuration
- Peripherals
- Logs
- Software information
- Room configuration

9.3 SSH and xAPI

SSH and xAPI are used for advanced administration, automation, and integration.

Use them only with authorized access and documented configuration procedures.

9.4 Control Hub

Control Hub supports remote monitoring and administration for Webex-managed devices.

It can be used for:

- Device status
- Diagnostics
- Logs
- Software upgrades
- Configuration
- Monitoring call quality
- Remote support

10. Normal Operation

During normal operation:

- Codec powers on automatically after power is applied.
- Power switch is on.
- Room Navigator connects to the codec.
- Display shows the room interface.
- Network connection is active.
- Device is registered to Webex, CUCM, or Expressway.
- Camera is detected and provides video.
- Microphones provide room audio.
- Speakers or display audio output works.
- USB-C content sharing works when connected.
- Wireless sharing works when enabled.
- Calls can be placed and received.
- Control Hub or management interface shows normal status.
- No ventilation or overheating warnings appear.

Useful operational checks:

```
Check display output
Check Room Navigator status
Check network IP address
Check registration status
Check camera preview
Check microphone audio levels
Check speaker output
Check content sharing
Check connected peripherals
Check RoomOS software version
Check Control Hub status if used
```

11. Troubleshooting

Troubleshooting should move from physical checks to system-level checks:

1. Power
2. Ventilation
3. Display cable and input
4. Camera connection
5. Microphone and audio wiring
6. Ethernet or Wi-Fi
7. Registration
8. Room Navigator
9. Content sharing

- 10. Peripherals and PoE budget
 - 11. RoomOS software
 - 12. Hardware fault
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11.1 Codec Does Not Power Up

Symptom:

- No display output
- Room Navigator inactive
- Codec appears off
- No boot behavior

Possible causes:

- AC power cable disconnected
- Power outlet failed
- Power switch off
- Wrong power cable
- Internal hardware fault
- Connected device power issue confused with codec power issue

Checks:

```
Check AC power outlet
Check power cable
Check codec power switch
Check connected display power
Check camera power if relevant
Observe boot behavior on display
```

Fix actions:

- Reseat AC power cable.
 - Turn on the codec power switch.
 - Test the outlet.
 - Replace power cable if faulty.
 - Check camera power separately.
 - Escalate if codec remains off with known-good power.
-

11.2 No Image on Display

Symptom:

- Display is blank
- Display shows no signal
- Codec appears powered but nothing is visible

Possible causes:

- Display powered off
- Wrong display input selected
- HDMI cable disconnected
- Bad HDMI cable
- Wrong HDMI output used
- Resolution mismatch
- Codec not fully booted

Checks:

```
Turn on display
Select correct HDMI input
Reseat HDMI cable
Try another HDMI cable
Try another HDMI output if appropriate
Check Room Navigator status
```

Fix actions:

- Select the correct HDMI input manually if needed.
- Replace HDMI cable.
- Confirm display supports the output resolution.
- Reboot codec only if cabling and display input are verified.

11.3 Camera Not Detected

Symptom:

- No camera image
- Camera unavailable in interface
- Calls have no local video
- Camera does not respond to control

Possible causes:

- HDMI camera not connected to input 1
- Camera not powered
- PoE budget exceeded
- Wrong cable
- Unsupported camera connection method
- VISCA cable used incorrectly
- Camera damaged
- RoomOS needs update

Checks:

```
Confirm camera HDMI is connected to input 1
Check camera power
Check PoE port and budget
Check Ethernet/IP peripheral connection if used
Check camera cable
Verify camera compatibility
```

Fix actions:

- Move HDMI camera to input 1.
- Power camera through supported PoE or specified power supply.
- Do not use a VISCA cable.
- Check PoE++ budget across ports 1–4.
- Update software if required.
- Replace cable or camera if confirmed faulty.

11.4 Camera Image Upside Down

Symptom:

- Camera image appears inverted
- Local or far-end video orientation is wrong

Possible cause:

- PTZ camera mounted upside down or inverted

Fix actions:

```
Open codec camera settings
Enable camera image flip/orientation correction
Save configuration
Test local preview and call video
```

11.5 Microphone or Audio Input Problem

Symptom:

- Far-end cannot hear room
- Audio level is low
- Microphone not detected
- Audio cuts in and out

Possible causes:

- Microphone cable disconnected

- Wrong audio input
- Peripheral not detected
- PoE/IP microphone issue
- Muted microphone
- Incorrect audio configuration
- Faulty microphone
- Room noise or placement issue

Checks:

```
Check microphone connection
Check mute state
Check audio input configuration
Check peripheral detection
Check PoE/network status if IP microphone
Test with known-good microphone if available
```

Fix actions:

- Reseat cable.
- Unmute.
- Verify audio input settings.
- Check microphone placement.
- Verify peripheral power and detection.
- Update software if necessary.

11.6 No Speaker or Output Audio

Symptom:

- Room cannot hear far end
- Content audio missing
- Volume control has no effect

Possible causes:

- Display audio muted
- Speaker system off
- Wrong audio output
- HDMI audio issue
- Analog audio cable issue
- Codec volume muted
- External DSP/audio system misconfigured

Checks:

```
Check codec volume
Check display volume
Check speaker amplifier or DSP
Check audio output configuration
Check HDMI or analog audio cabling
Test call audio and content audio separately
```

Fix actions:

- Unmute codec/display/speaker system.
- Select correct audio output.
- Reseat cables.
- Check external audio processor settings.
- Verify content source audio.

11.7 Endpoint Offline or Not Registered

Symptom:

- Device cannot make or receive calls
- Control Hub/CUCM/Expressway shows offline
- Registration warning appears
- Calls fail

Possible causes:

- Ethernet disconnected
- Wi-Fi unavailable or not supported
- Wrong VLAN
- DHCP failure
- DNS issue
- Firewall blocks required traffic
- Registration credentials issue
- Time/certificate issue
- Service outage

Checks:

```
Check Ethernet or Wi-Fi connection
Check IP address
Check default gateway
Check DNS
Check registration status
Check time and date
Check firewall rules
Check platform status
```

Fix actions:

- Restore network connectivity.
 - Correct VLAN or IP settings.
 - Fix DNS/gateway.
 - Re-enter registration settings if needed.
 - Update certificates/time settings.
 - Escalate to network or collaboration admin if external service is blocked.
-

11.8 Room Navigator Not Working

Symptom:

- Touch controller does not connect
- Room controls unavailable
- Device cannot be controlled from room interface

Possible causes:

- Room Navigator not powered
- Network cable disconnected
- Wrong pairing
- PoE issue
- VLAN or discovery issue
- Software mismatch
- Codec not online

Checks:

```
Check Room Navigator cable
Check PoE or power
Check pairing status
Check network path to codec
Check codec status
```

Fix actions:

- Reseat cable.
 - Verify PoE.
 - Re-pair Room Navigator if needed.
 - Confirm both devices are on the correct network.
 - Update software if required.
-

11.9 Peripheral Not Detected

Symptom:

- Camera, microphone, or USB device not visible
- Peripheral does not respond
- Room feature unavailable

Possible causes:

- Cable disconnected
- PoE budget exceeded
- USB device unsupported
- Wrong port
- Peripheral needs firmware update
- Faulty peripheral

Checks:

```
Check cable
Check port
Check PoE++ budget
Check peripheral compatibility
Check Control Hub or web interface diagnostics
```

Fix actions:

- Reseat cable.
- Use supported port.
- Reduce PoE load if needed.
- Update software.
- Test with known-good peripheral.

11.10 Content Sharing Does Not Work

Symptom:

- Laptop content not displayed
- USB-C sharing unavailable
- Wireless share fails

Possible causes:

- Wrong cable
- Laptop USB-C does not support video output
- Wireless sharing disabled
- Miracast/AirPlay blocked by network
- Client device incompatible
- Wrong display layout
- Software issue

Checks:

```
Check USB-C cable
Check laptop video support
Check sharing settings
Check wireless sharing feature
Check network discovery requirements
Check display output
```

Fix actions:

- Use a video-capable USB-C cable.
- Enable wireless sharing where required.
- Check network/firewall rules.
- Use Webex sharing if available.
- Update RoomOS if needed.

11.11 Overheating or Shutdown

Symptom:

- Device shuts down
- Warnings appear
- Room system becomes unstable
- Codec is hot

Possible causes:

- Ventilation blocked
- Less than required clearance
- Rack temperature too high
- Dust buildup
- Equipment stacked incorrectly

Checks:

```
Check front and rear clearance
Confirm at least 10 cm / 4 in. airflow clearance
Check rack temperature
Check vents for obstruction
```

Fix actions:

- Clear ventilation openings.
- Improve rack airflow.
- Move equipment if stacked incorrectly.
- Keep room/rack within operating temperature.
- Clean dust carefully.

11.12 Software or RoomOS Issue

Symptom:

- Unexpected behavior
- Peripheral compatibility issue
- Known bug suspected
- Features missing
- System unstable

Possible causes:

- Outdated RoomOS
- Failed update
- Unsupported peripheral firmware
- Configuration issue
- Registration platform mismatch

Fix actions:

```
Check current RoomOS version
Review update availability
Upgrade through supported Cisco method
Reboot after update if required
Verify all peripherals after update
Collect logs before escalation
```

12. Factory Reset and Recovery

Factory reset and recovery procedures are documented in the Cisco Board, Desk, and Room Series Administration Guide for RoomOS endpoints.

Factory reset may be used when:

- Device must be returned to default settings
- Configuration is corrupted
- Device is being repurposed
- Registration must be cleared
- Recovery is required after serious configuration issues

Important warning:

A factory reset removes local configuration and requires the codec to be re-provisioned afterward.

Before reset:

```
Document current registration method
Back up or record configuration where possible
Record IP settings
Record room device connections
Confirm re-provisioning information is available
Plan downtime
```

After reset:

```
Complete setup wizard
Reconnect to network
Register to Webex, CUCM, or Expressway
Reconfigure peripherals
Test displays, camera, microphones, calls, and sharing
```

Use the official RoomOS administration procedure for the exact software version before performing a reset.

13. Maintenance and Best Practices

Recommended practices:

- Keep RoomOS software current.
- Use Control Hub for monitoring when deployed.
- Keep a record of registration method and device ID.
- Label HDMI, Ethernet, USB, and audio cables.
- Keep at least 10 cm / 4 in. clearance at front and rear.
- Do not block vents.
- Keep rack temperature within supported range.
- Use compatible cameras, microphones, and peripherals.
- Do not exceed PoE++ budget.
- Avoid unsupported cables such as VISCA for camera control.
- Test audio, video, and sharing after changes.
- Document display input selection.
- Monitor endpoint status regularly.
- Collect logs before escalation.
- Power-cycle only when required.
- Plan software upgrades during maintenance windows.

Useful maintenance checks:

```
Check registration status
Check network status
Check RoomOS version
Check camera preview
Check microphone status
```

- Check display output
- Check Room Navigator
- Check peripheral list
- Check wireless sharing status
- Check Control Hub alerts
- Check ventilation clearance

14. Quick Reference

Power Check

- Check AC power cable
- Check outlet
- Check codec power switch
- Check camera power separately
- Check display power

Network Check

- Verify Ethernet or Wi-Fi
- Confirm IP address
- Check gateway and DNS
- Check registration status
- Confirm Webex/CUCM/Expressway reachability

Display Check

- Turn on display
- Select correct HDMI input
- Check HDMI cable
- Check codec HDMI output
- Check resolution compatibility

Camera Check

- Connect HDMI camera to input 1
- Check camera power
- Check PoE budget
- Do not use VISCA cable
- Flip image in settings if mounted inverted

Audio Check

- Check microphone connection
- Check mute state
- Check speaker/display volume
- Check audio input/output settings
- Check external DSP if used

Peripheral Check

- Check Ethernet/USB cable
- Check PoE++ budget
- Check Control Hub or web diagnostics
- Verify compatibility

Content Sharing Check

- Check USB-C cable
- Confirm laptop supports video output
- Verify wireless sharing settings
- Check Miracast/AirPlay/Webex sharing support

Management Access

- Use Room Navigator
- Use HTTPS web interface
- Use SSH where authorized
- Use xAPI for integration
- Use Control Hub where deployed

Ventilation

- Maintain 10 cm / 4 in. clearance at front and rear
- Do not block vents
- Keep rack temperature within limits

Restart

- Power-cycle only when required
- Use management interface where possible
- Verify normal operation after restart

Factory Reset / Recovery

Use official RoomOS Administration Guide procedure
Back up or document configuration first
Re-provision after reset
Test all room functions
